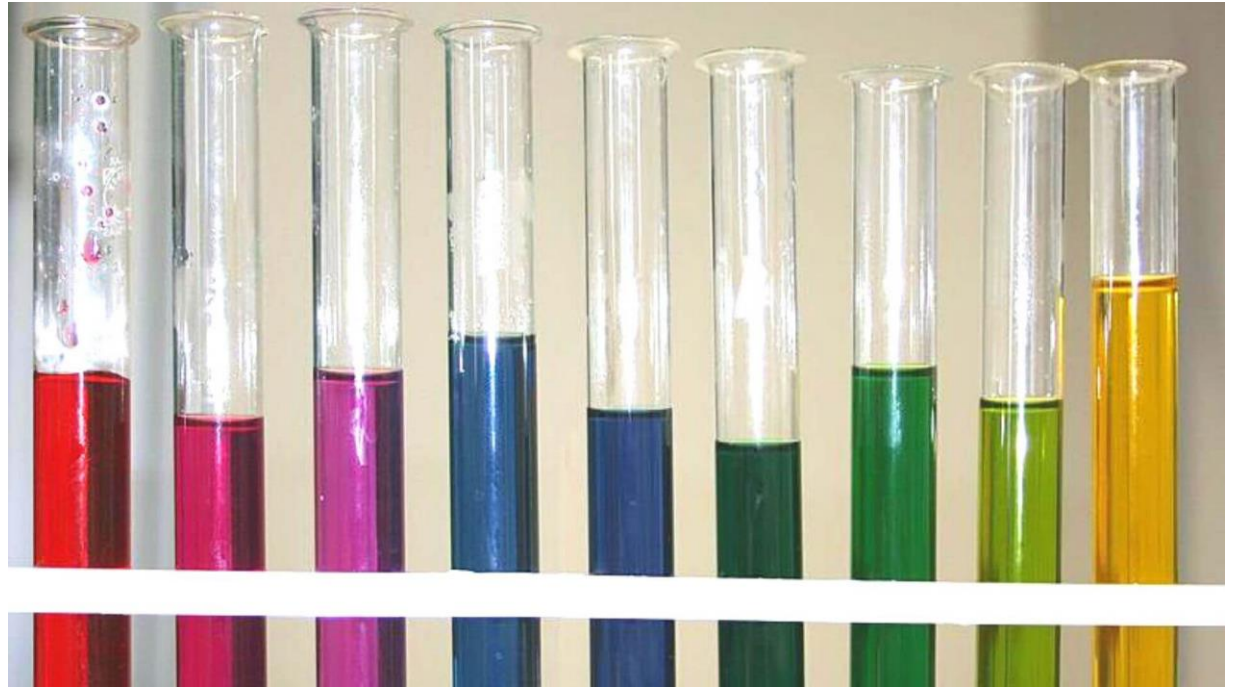


Controller Tuning to Adjust the pH of a Solution

Nicole Gorostiza
Peihan Liu
Gretna Truong
Brian Chu



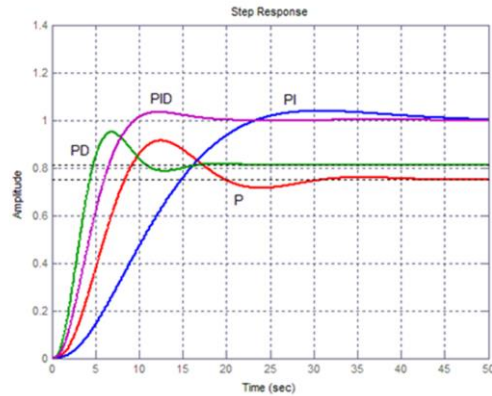
Jacobs School of Engineering
UCSD

16 May 2023

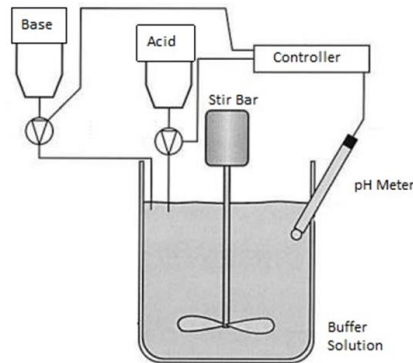


[osc.org]

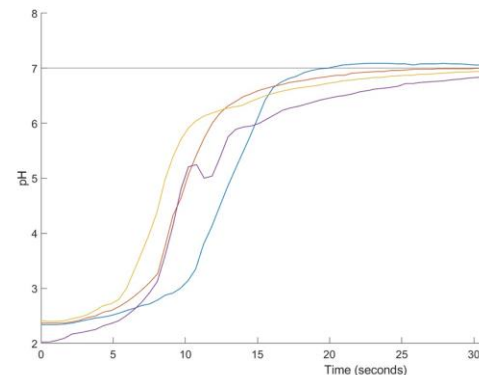
This presentation focuses on finding the best controller and tuning for pH control



pH control uses and types of controllers

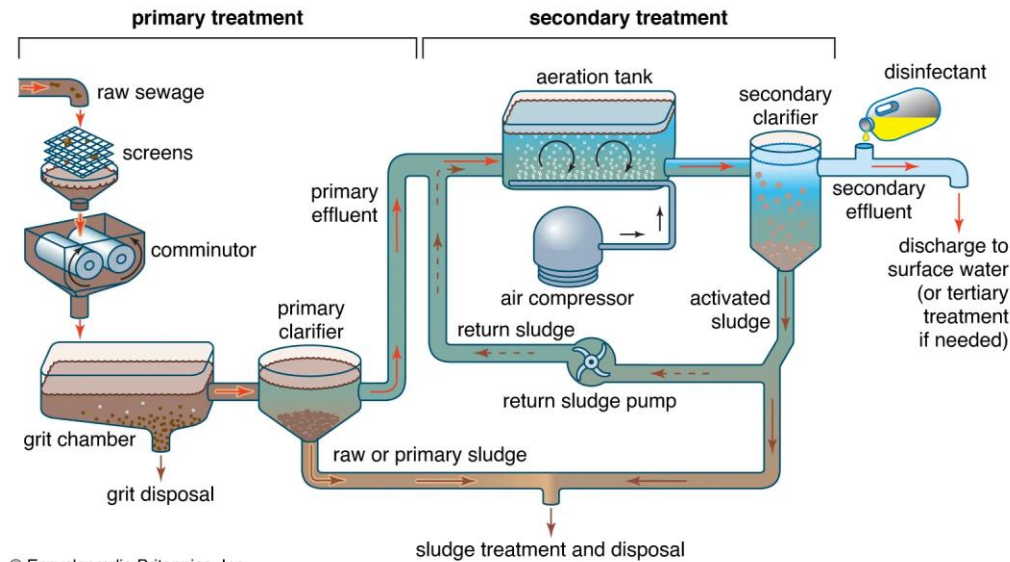


Experimental set up



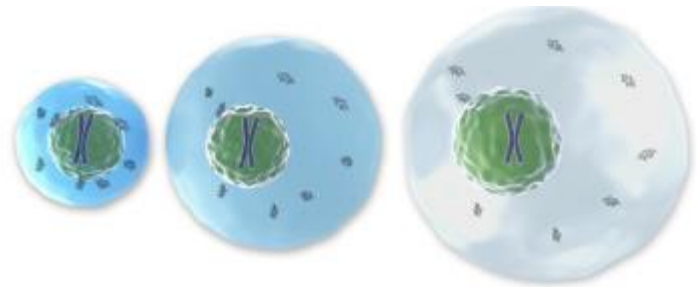
Results

pH control is useful in wastewater treatment and cell growth rate



Acidic or basic water
must be neutralized

Controlled pH cell
environments allows for
consistent data



Various types of controller modes can be used for different systems

P-Only:

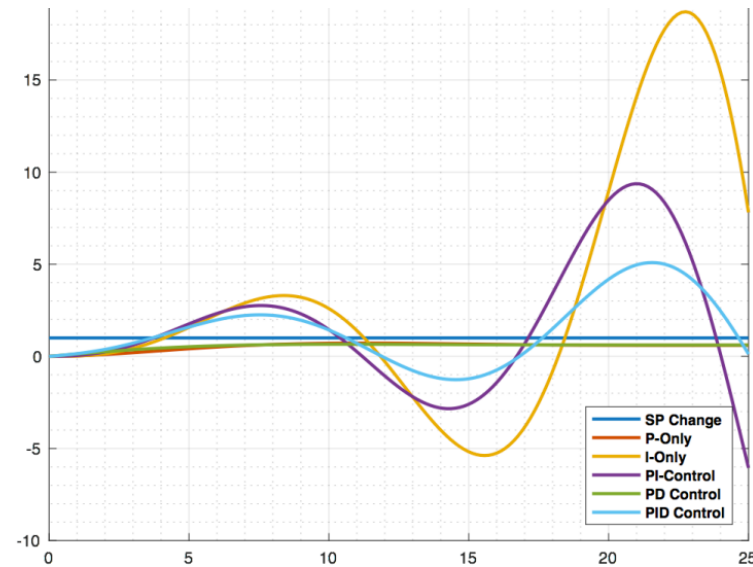
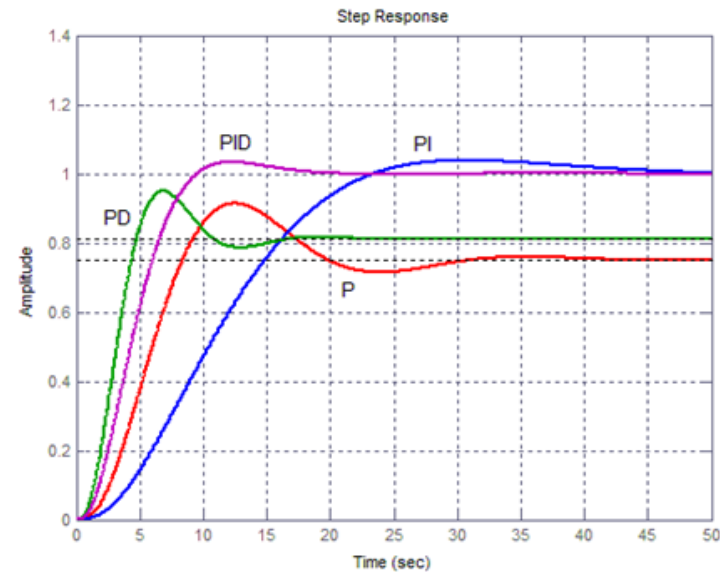
Action is immediate

I-Only:

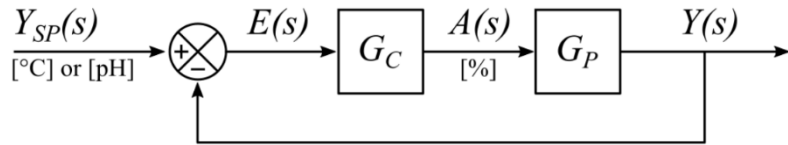
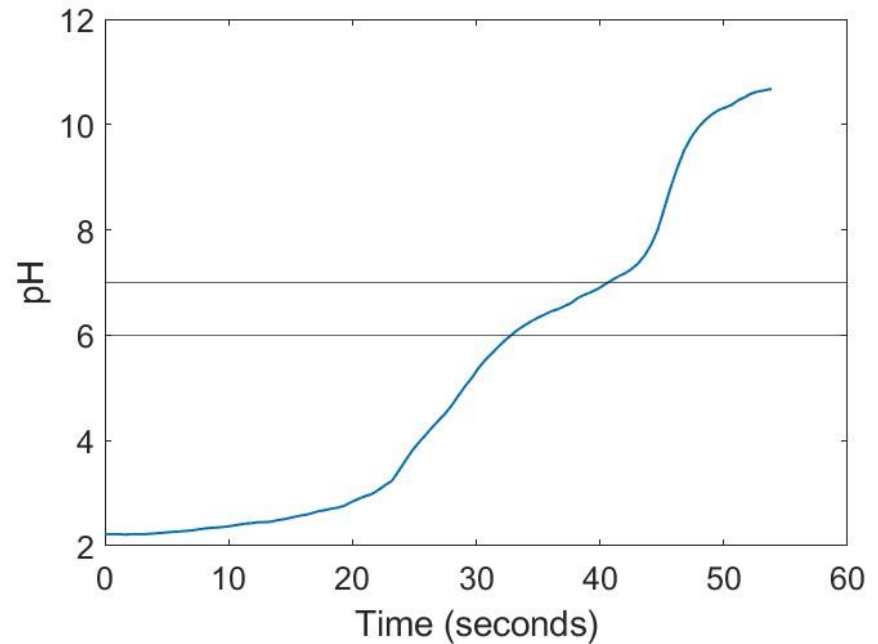
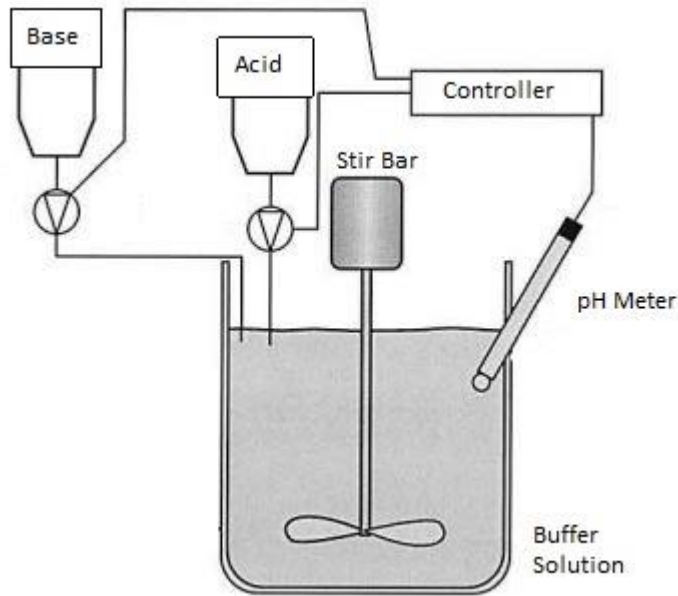
Eliminates steady-state offset

D-Only:

Has stabilizing effect



The experiment was conducted using .1 M HCl, .1 M NaOH, PBS, and a pump with a controller



Process model

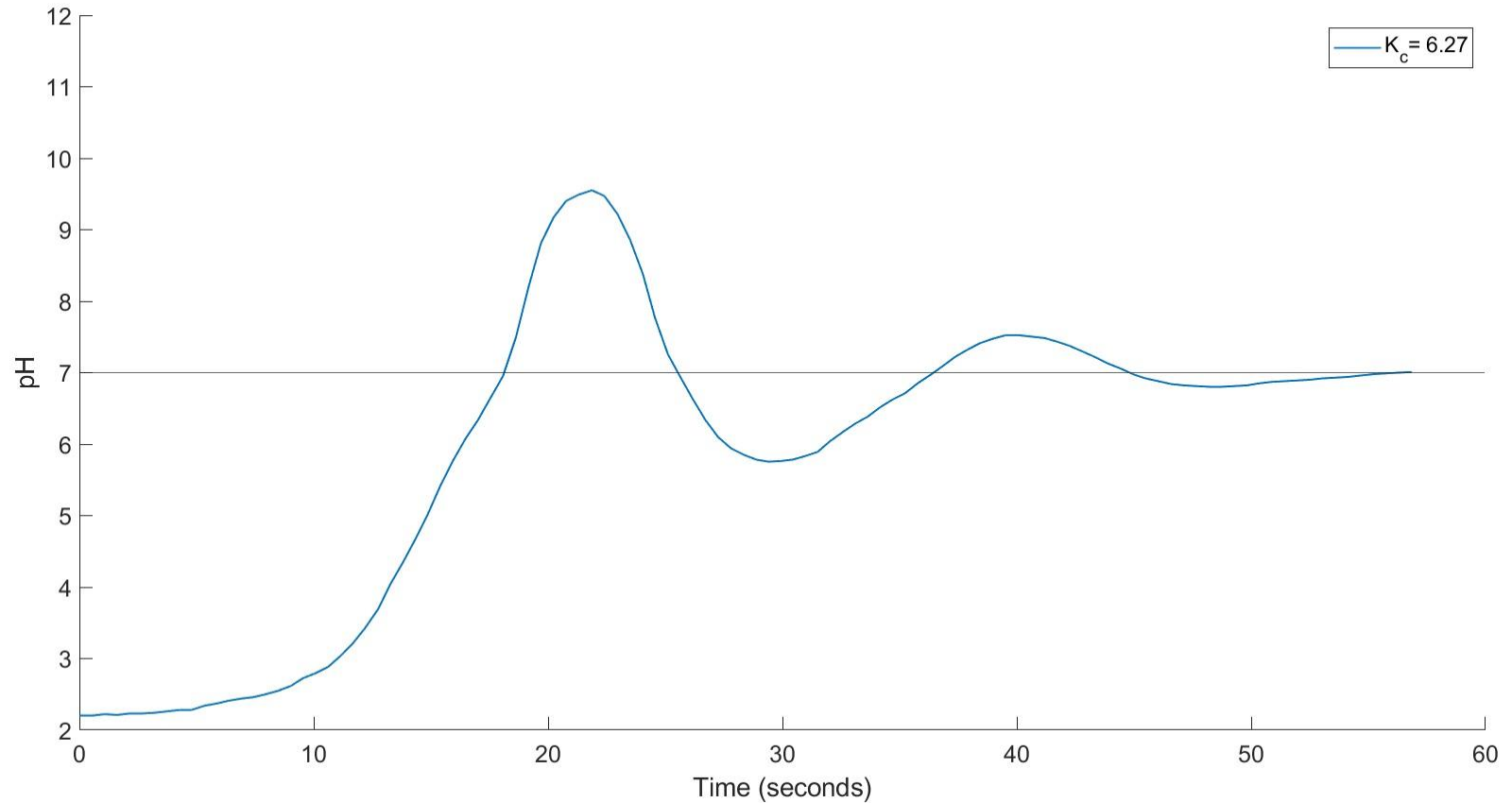
$$\frac{K_p}{s(\tau_p s + 1)}$$

Controller Model

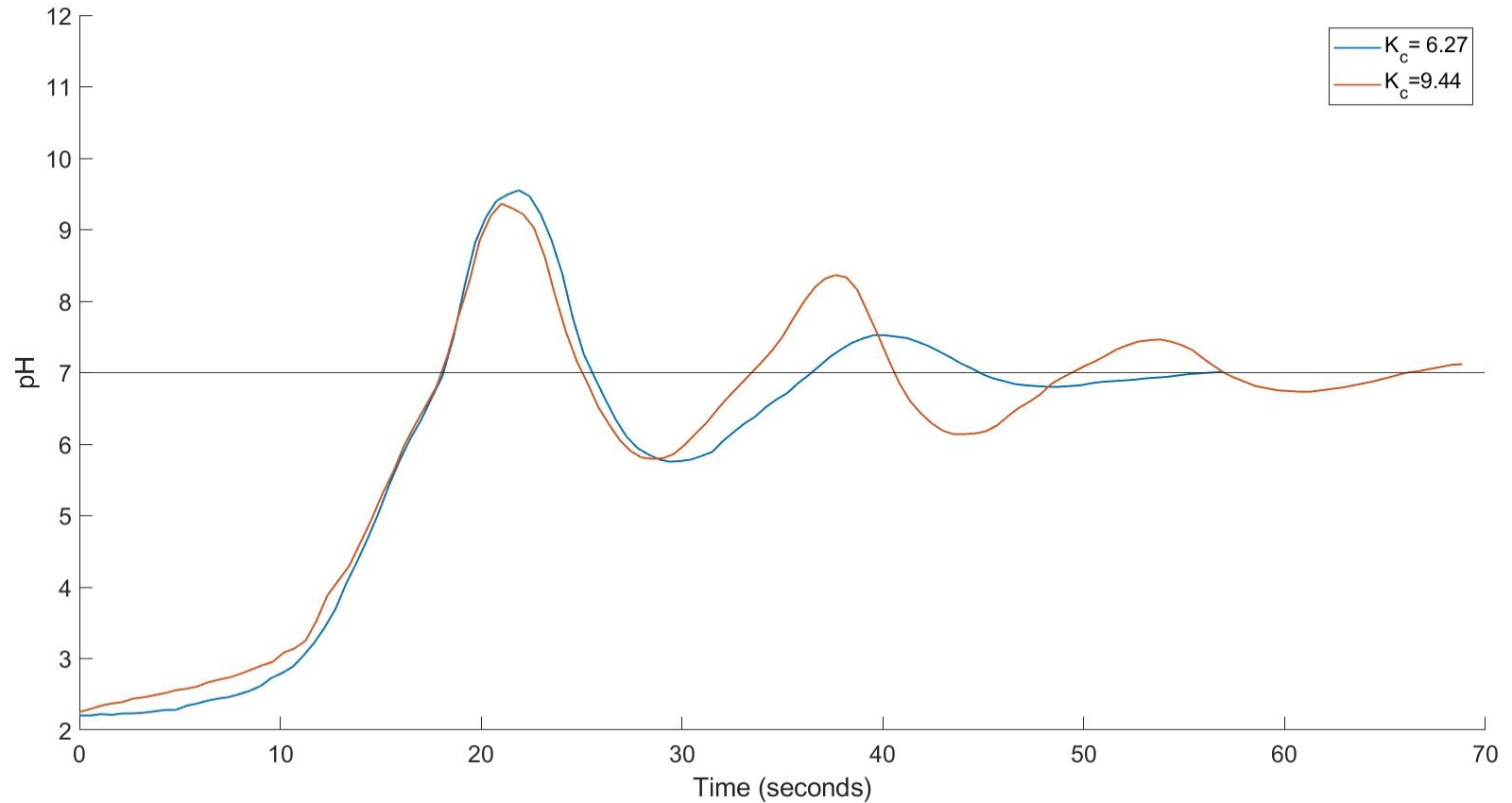
$$K_c = (1 + \tau_d s)$$

$$K_c = \frac{1}{K_p \tau_c} \quad \tau_d = \tau_p$$

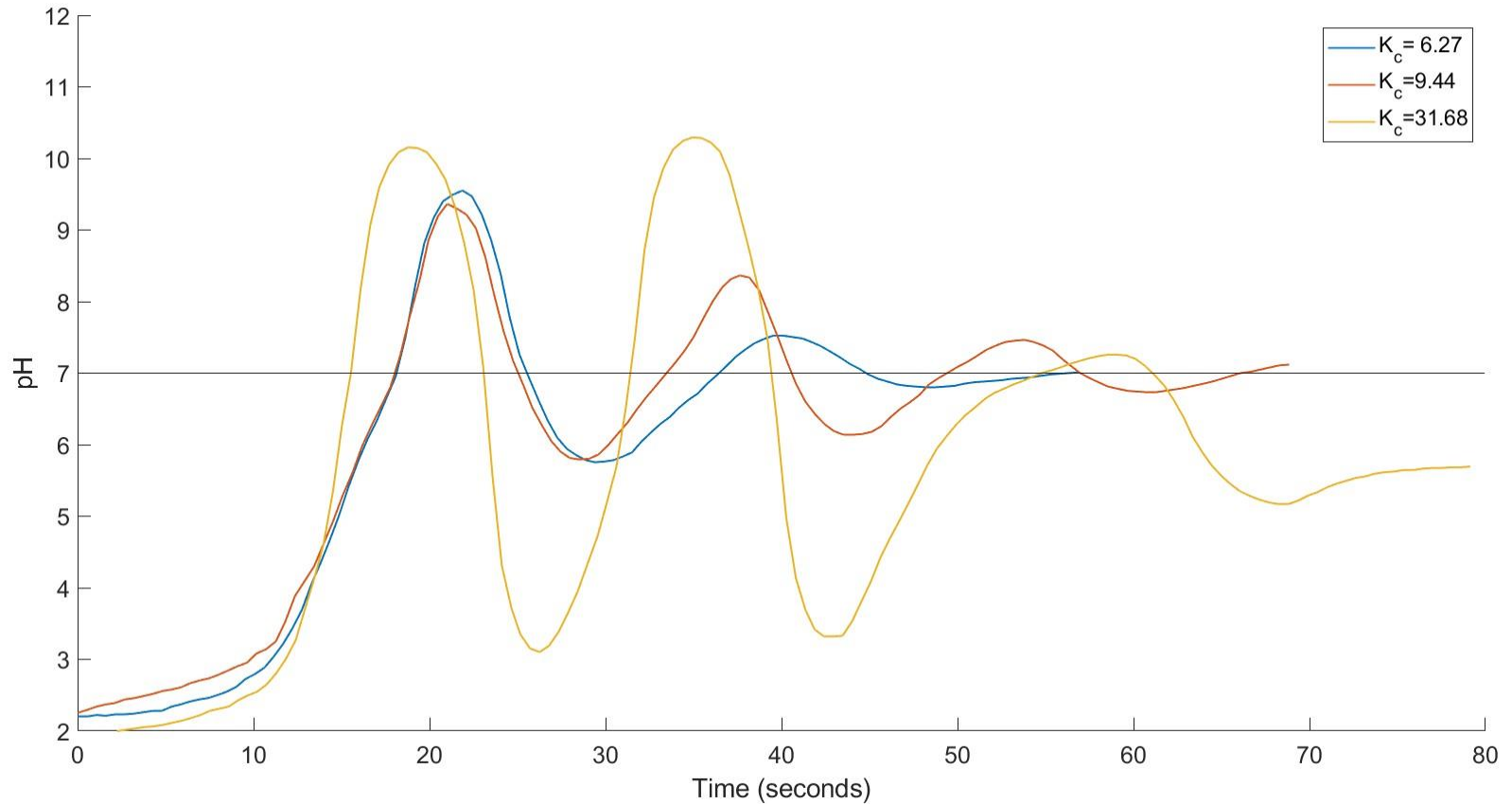
P control creates large offshoots but quickly stabilizes



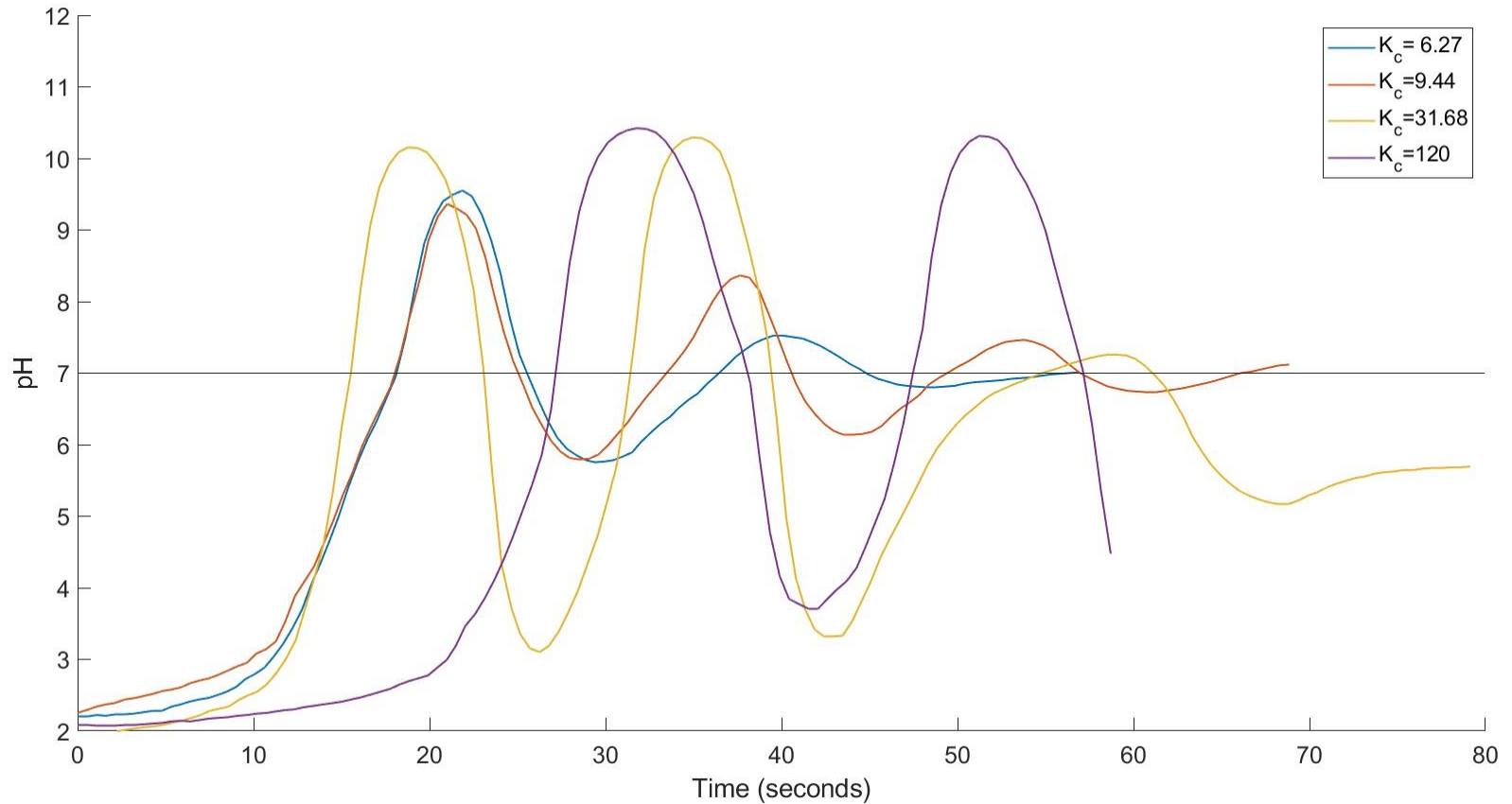
P control creates large offshoots but quickly stabilizes



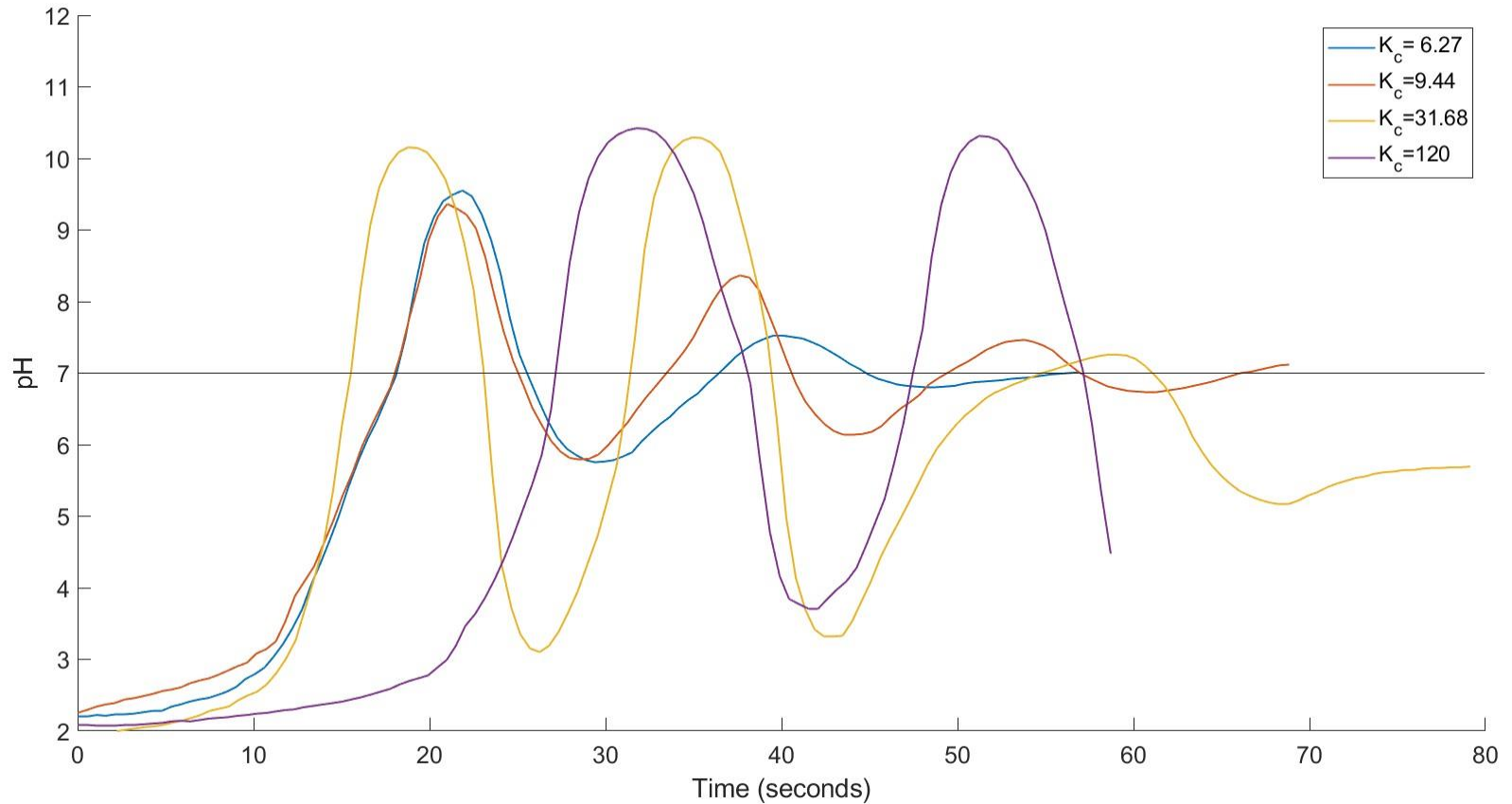
P control creates large offshoots but quickly stabilizes



P control creates large offshoots but quickly stabilizes

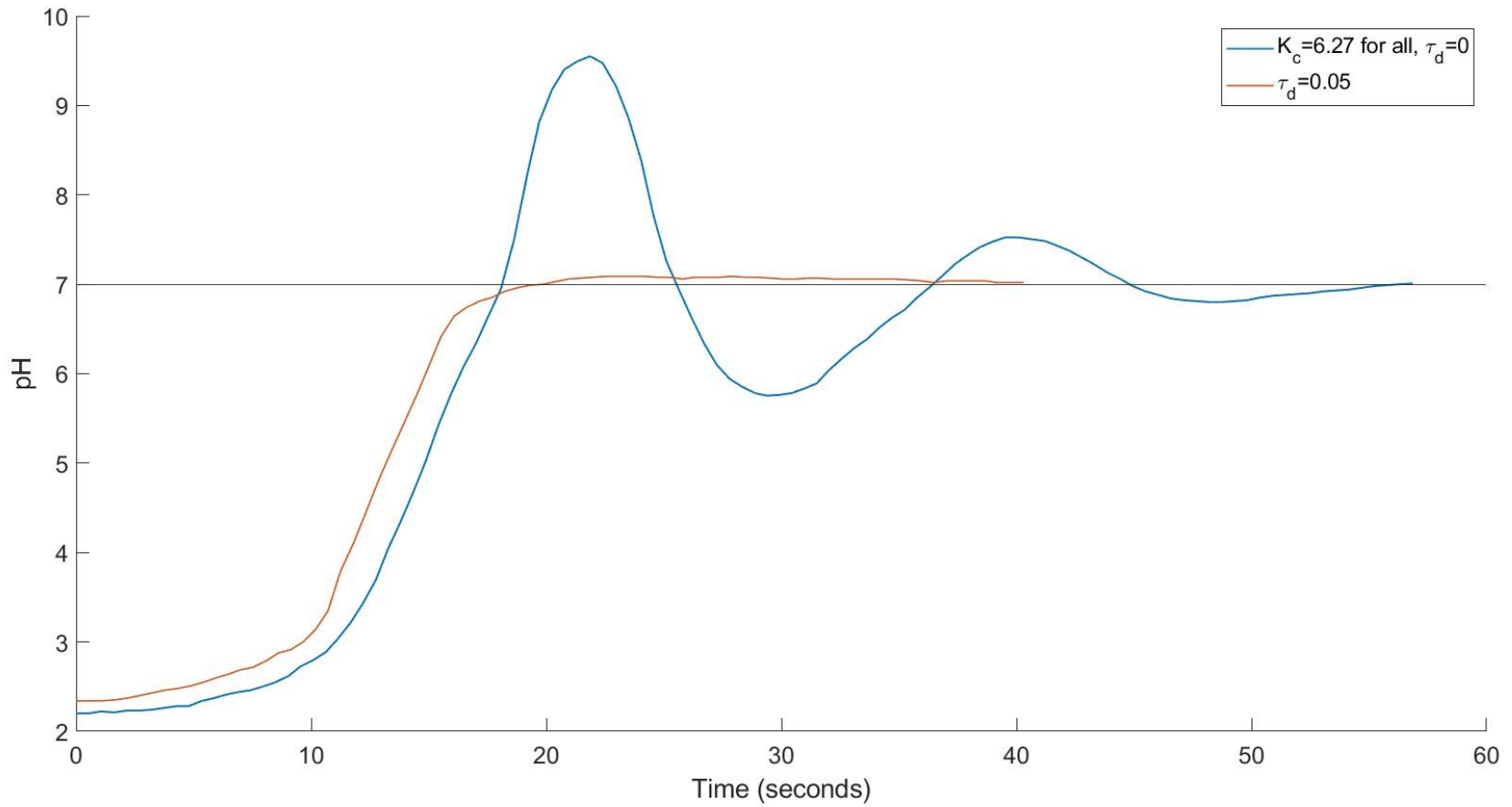


P control creates large offshoots but quickly stabilizes

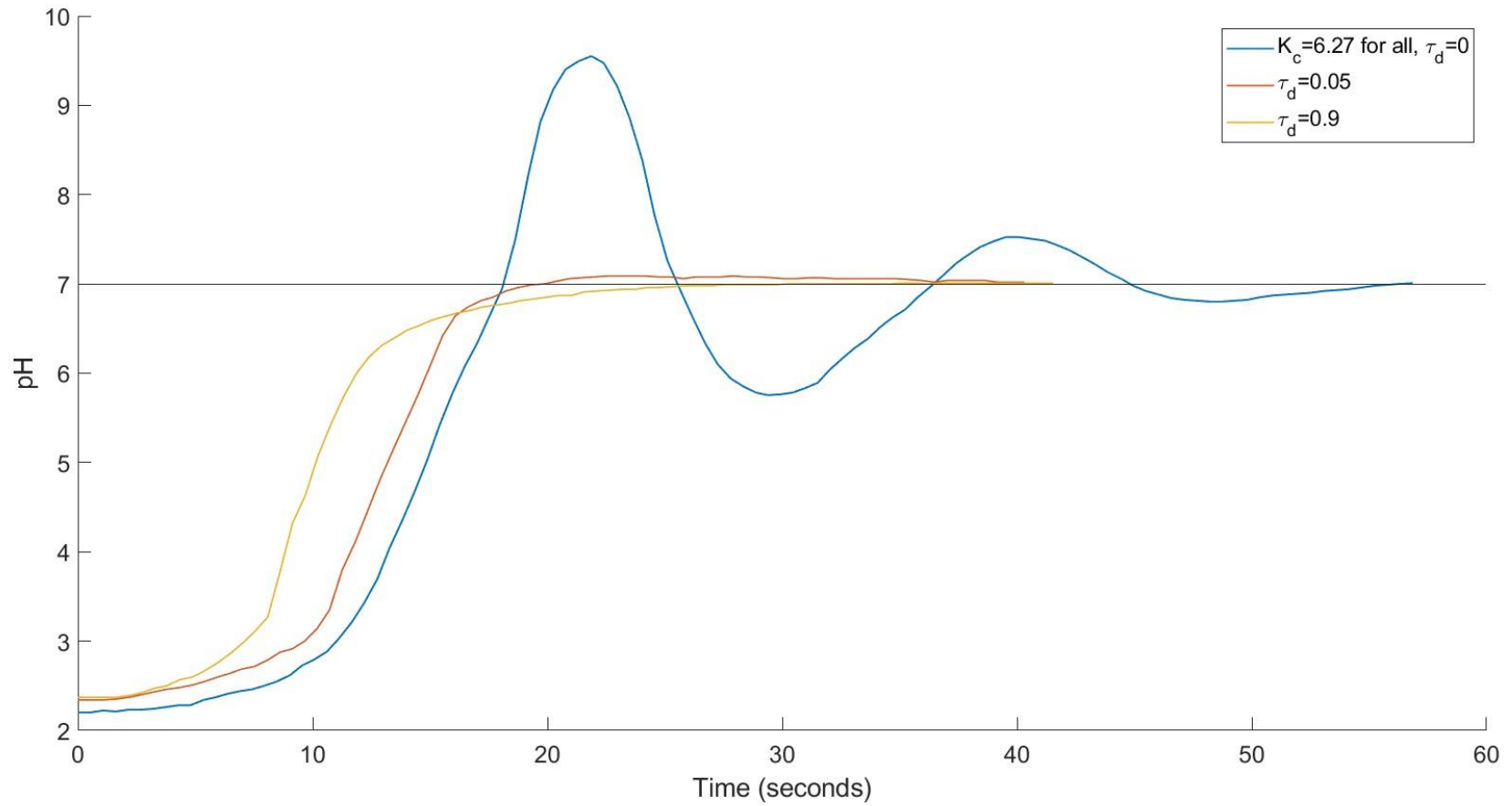


Too large of K_c value creates oscillation

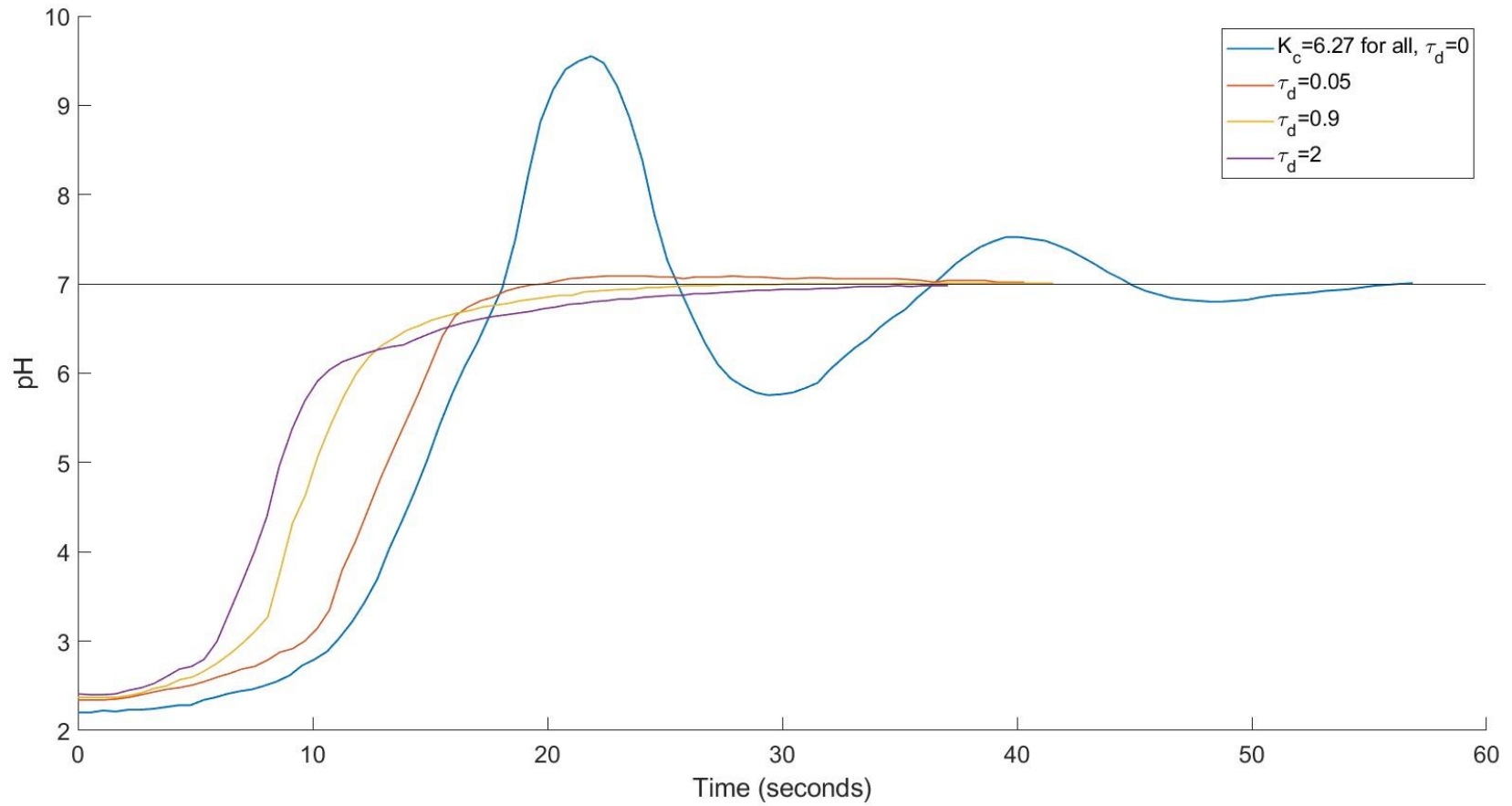
PD control eliminates offshoot



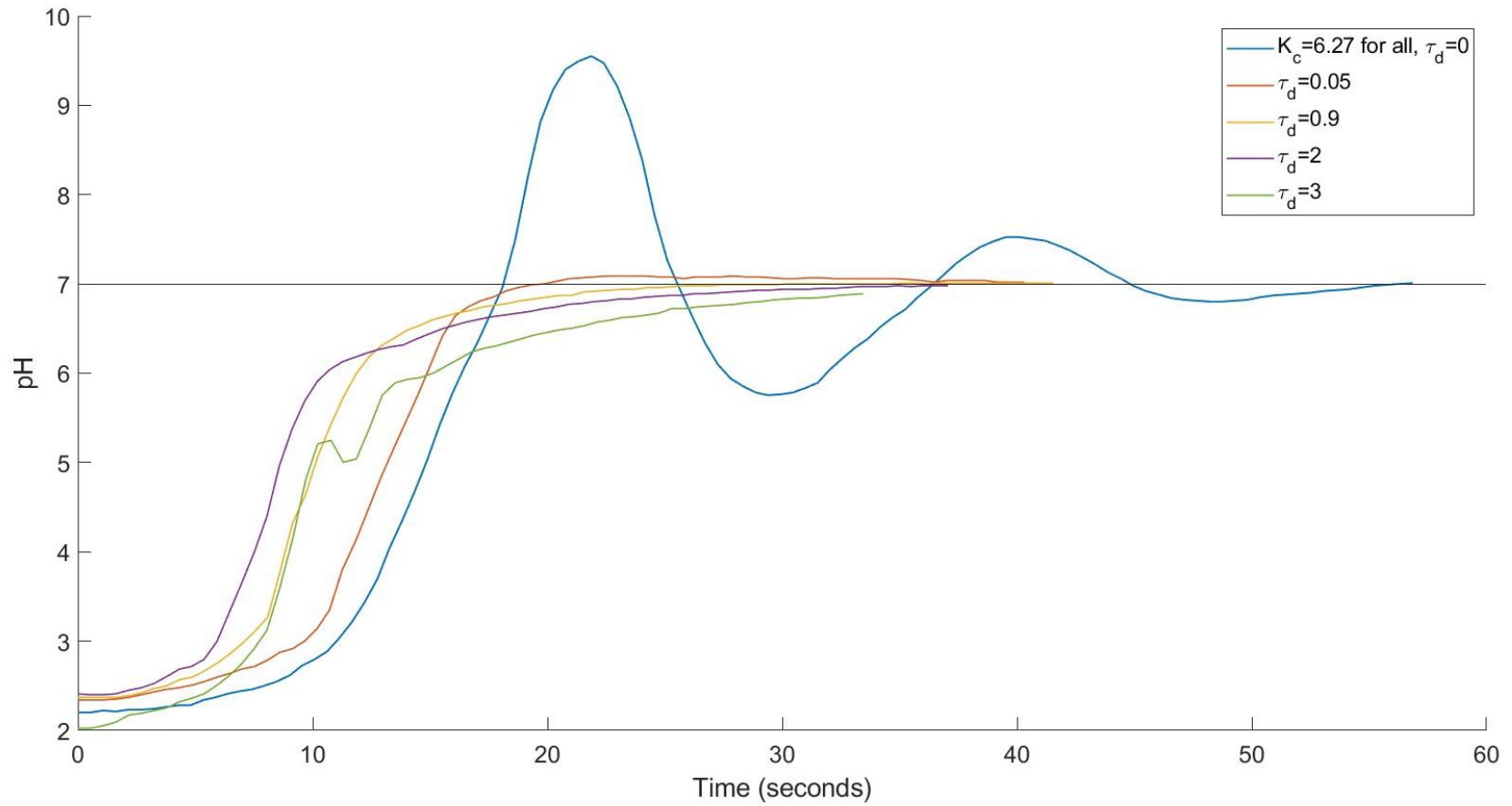
PD control eliminates offshoot



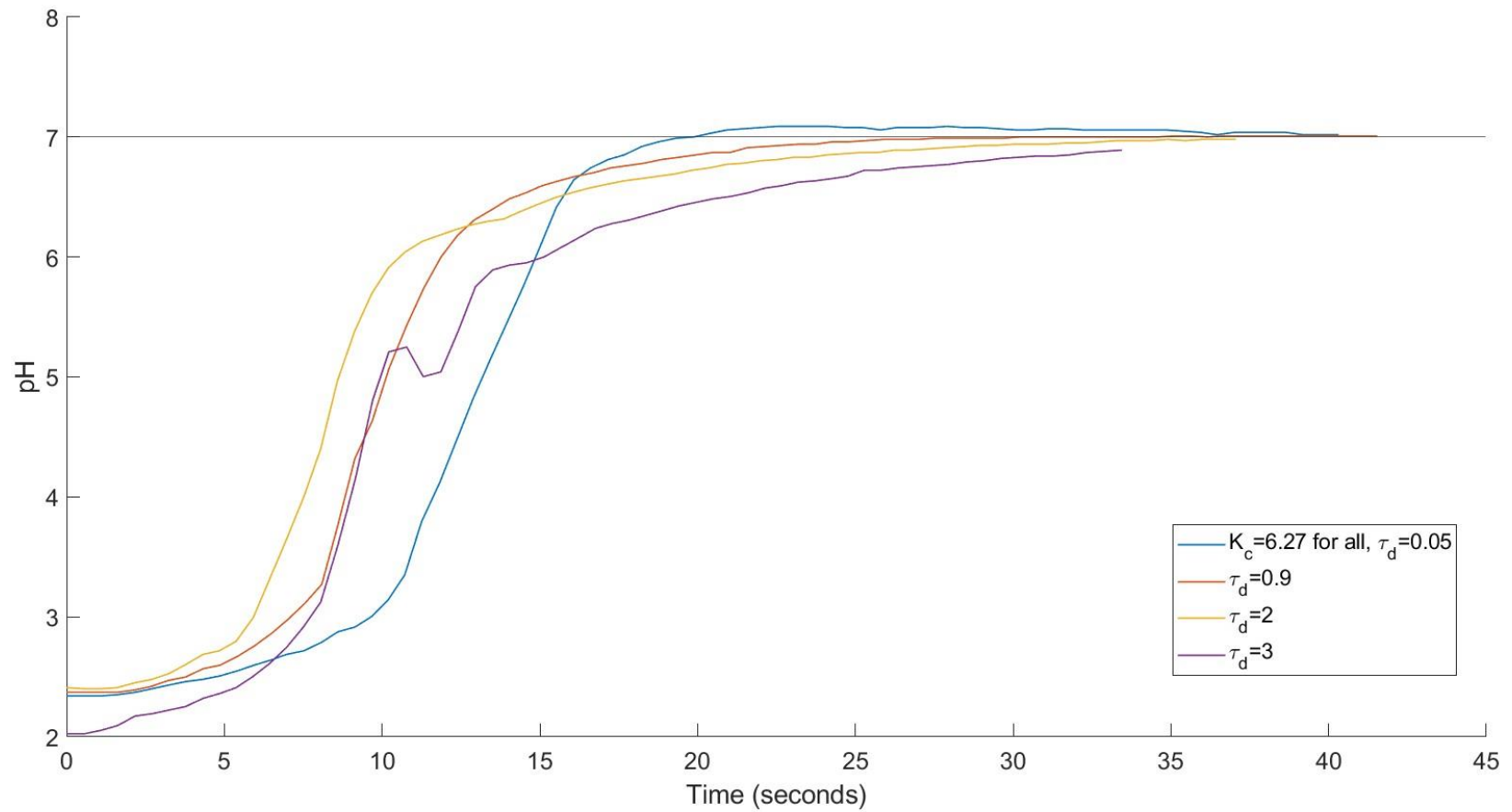
PD control eliminates offshoot



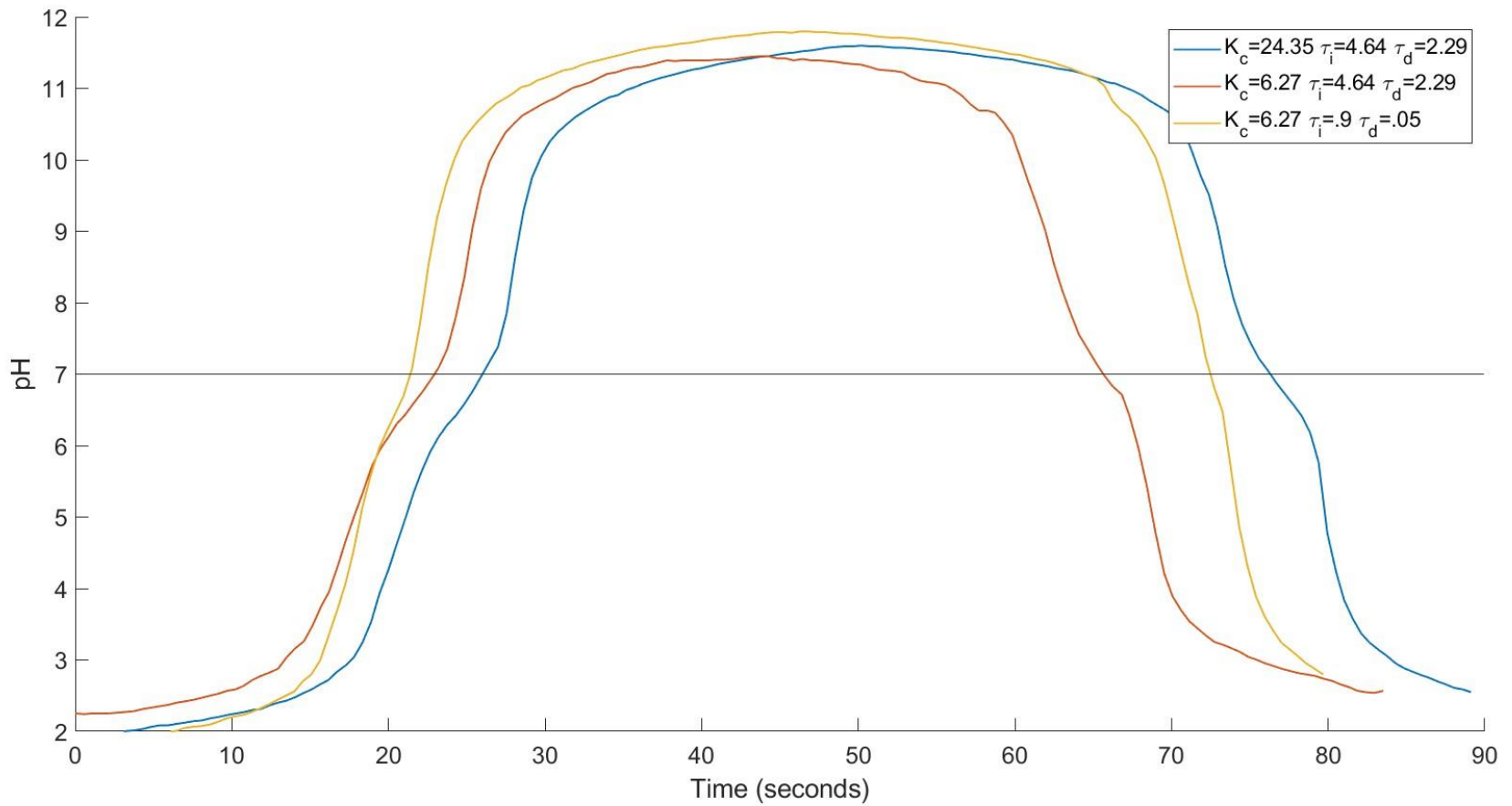
PD control eliminates offshoot



PD control with $\tau=0.9$ creates best results



PID control makes system unstable

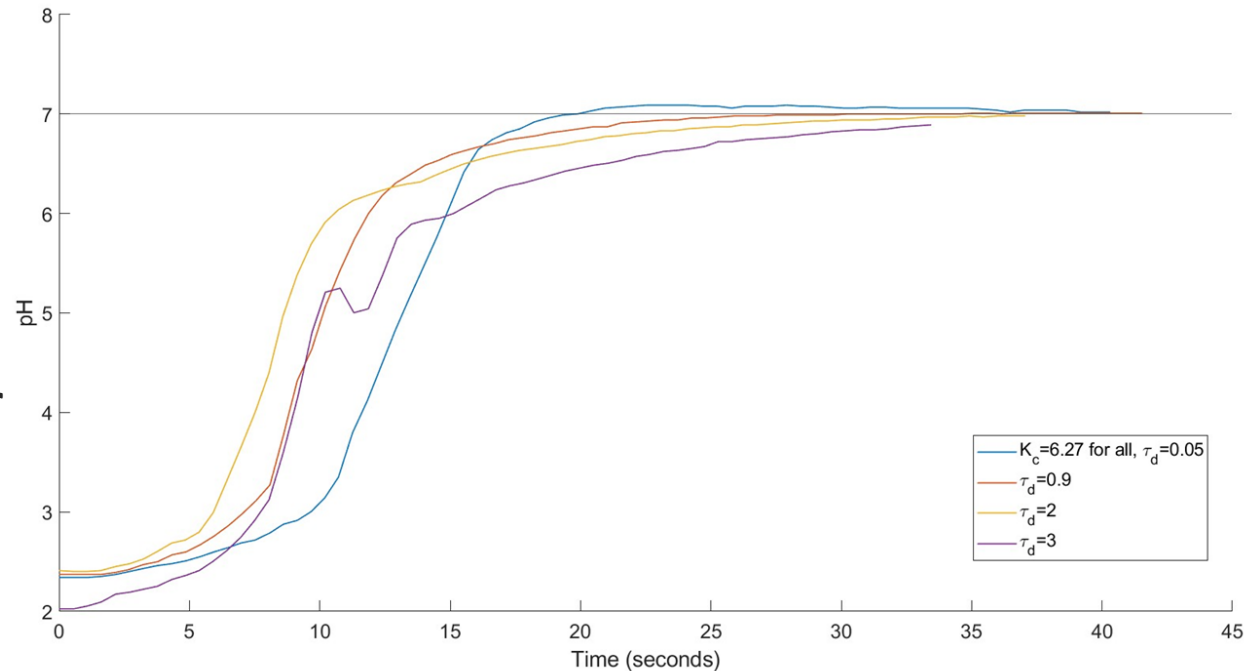


In summary, the experimental results verified the direct synthesis results since PD control was the best

The best parameters were found to be $K_c=6.25$ and $\tau_d=0.9$



Having the wrong controller type can significantly impact the system



Questions?